Q1.

A student states that $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\cos x - \sin x}\,dx$ is not an improper integral because $\dfrac{\cos x + \sin x}{\cos x - \sin x}$ is defined at both $x = 0$ and $x = \dfrac{\pi}{2}$

Assess the validity of the student's argument.

(Total 2 marks)

Q2.

Evaluate the improper integral $\displaystyle\int_0^{\infty} \frac{4x - 30}{(x^2 + 5)(3x + 2)}\,dx$, showing the limiting process used.

Give your answer as a single term.

(Total 8 marks)

Q3.

Show that the improper integral $\displaystyle\int_{25}^{\infty} \frac{1}{x\sqrt{x}}\,dx$ has a finite value and find that value.

(Total 4 marks)

Q4.

(a) Explain why $\displaystyle\int_0^1 x^4 \ln x\,dx$ is an improper integral.

(1)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(b) Evaluate $\displaystyle\int_0^1 x^4 \ln x\,dx$, showing the limiting process used.

(6)

(Total 7 marks)

Q5.

For the improper integral $\displaystyle\int_1^{\infty} x^{-4}(2x^2 - 5x)\,dx$, either show that the integral has a finite value and state its value, or explain why the integral does not have a finite value.

(Total 3 marks)

Q6.

Evaluate the improper integral

$$\int_0^{\infty} \left(\frac{2x}{x^2 + 4} - \frac{4}{2x + 3}\right) dx$$

For more help, please visit our website www.exampaperspractice.co.uk

showing the limiting process used and giving your answer in the form ln k, where k is a constant.

(Total 6 marks)

Q7.

Show that only one of the following improper integrals has a finite value, and find that value:

(a) $\int_8^{\infty} x^{-\frac{2}{3}}\,dx$;

(b) $\int_8^{\infty} x^{-\frac{4}{3}}\,dx$.

(Total 5 marks)

Q8.

(a) Explain why $\int_{\frac{1}{2}}^{\infty} \frac{x(1-2x)}{x^2+3e^{4x}}\,dx$ is an improper integral.

(1)

(b) By using the substitution $u = x^2e^{-4x} + 3$, find

$$\int \frac{x(1-2x)}{x^2+3e^{4x}}\,dx$$

(3)

(c) Hence evaluate $\int_{\frac{1}{2}}^{\infty} \frac{x(1-2x)}{x^2+3e^{4x}}\,dx$, showing the limiting process used.

EXAM PAPERS PRACTICE

(4)

(Total 8 marks)

©2025 Exam Papers Practice. All rights reserved.

Q9.

(a) Find $\int x^2e^{-x}\,dx$.

(4)

(b) Hence evaluate $\int_0^{\infty} x^2e^{-x}\,dx$, showing the limiting process used.

(3)

(Total 7 marks)

Q10.

(a) Find, in terms of p and q, the value of the integral $\int_p^q \frac{2}{x^3}\,dx$.

(3)

(b) Show that only one of the following improper integrals has a finite value, and find

For more help, please visit our website www.exampaperspractice.co.uk

that value:

(i) $\int_0^2 \frac{2}{x^3}\,dx$;

(ii) $\int_0^\infty \frac{2}{x^3}\,dx$.

(3)
(Total 6 marks)

Q11.

(a) Write $\frac{4}{4x+1} - \frac{3}{3x+2}$ in the form $\frac{C}{(4x+1)(3x+2)}$, where C is a constant.

(1)

(b) Evaluate the improper integral

$$\int_1^\infty \frac{10}{(4x+1)(3x+2)}\,dx$$

showing the limiting process used and giving your answer in the form $\ln k$, where k is a constant.

(6)
(Total 7 marks)

Q12.

(a) Find $\int x^2 \ln x\,dx$.

(3)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(b) Explain why $\int_0^e x^2 \ln x\,dx$ is an improper integral.

(1)

(c) Evaluate $\int_0^e x^2 \ln x\,dx$, showing the limiting process used.

(3)
(Total 7 marks)

Q13.

(a) Explain why $\int_0^{\frac{1}{16}} x^{-\frac{1}{2}}\,dx$ is an improper integral.

(1)

(b) For each of the following improper integrals, find the value of the integral **or** explain briefly why it does not have a value:

For more help, please visit our website www.exampaperspractice.co.uk

(i) $\int_0^{\frac{1}{16}} x^{-\frac{1}{2}}\, dx$;

(3)

(ii) $\int_0^{\frac{1}{16}} x^{-\frac{5}{4}}\, dx$.

(3)
(Total 7 marks)

Q14.

(a) Explain why $\int_1^{\infty} \frac{\ln x^2}{x^3} dx$ is an improper integral.

(1)

(b) (i) Show that the substitution $y = \frac{1}{x}$ transforms $\int \frac{\ln x^2}{x^3} dx$ into $\int 2y \ln y\, dy$.

(2)

(ii) Evaluate $\int_0^1 2y \ln y\, dy$, showing the limiting process used.

(5)

(iii) Hence write down the value of $\int_1^{\infty} \frac{\ln x^2}{x^3} dx$.

(1)
(Total 9 marks)

EXAM PAPERS PRACTICE

Q15.

(a) Explain why $\int_1^{\infty} 4xe^{-4x}\, dx$ is an improper integral.

© 2025 Exam Papers Practice. All rights reserved.

(1)

(b) Find $\int 4xe^{-4x}\, dx$.

(3)

(c) Hence evaluate $\int_1^{\infty} 4xe^{-4x}\, dx$, showing the limiting process used.

(3)
(Total 7 marks)

Q16.

For each of the following improper integrals, find the value of the integral **or** explain why it does not have a value:

(a) $\int_1^{\infty} x^{-\frac{1}{4}}\, dx$;

For more help, please visit our website www.exampaperspractice.co.uk

(3)

(b) $\int_1^{\infty} x^{-\frac{5}{4}}\,dx$;

(3)

(c) $\int_1^{\infty} (x^{-\frac{3}{4}} - x^{-\frac{5}{4}})\,dx$.

(1)
(Total 7 marks)

Q17.

(a) Use integration by parts to show that $\int \ln x\,dx = x \ln x - x + c$, where c is an arbitrary constant.

(2)

(b) Hence evaluate $\int_0^1 \ln x\,dx$, showing the limiting process used.

(4)
(Total 6 marks)

Q18.

Evaluate the improper integral

$$\int_1^{\infty} \left(\frac{1}{x} - \frac{4}{4x+1}\right) dx$$

showing the limiting process used and giving your answer in the form $\ln k$, where k is a constant to be found.

(Total 5 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q19.

(a) Explain why $\int_1^{\infty} xe^{-3x}\,dx$ is an improper integral.

(1)

(b) Find $\int xe^{-3x}\,dx$.

(3)

(c) Hence evaluate $\int_1^{\infty} xe^{-3x}\,dx$, showing the limiting process used.

(3)
(Total 7 marks)

Q20.

For each of the following improper integrals, find the value of the integral **or** explain briefly why it does not have a value:

For more help, please visit our website www.exampaperspractice.co.uk

(a) $\int_0^\infty \frac{1}{\sqrt{x}}\,dx$;

(3)

(b) $\int_9^\infty \frac{1}{x\sqrt{x}}\,dx$.

(4)
(Total 7 marks)

Q21.

(a) Find $\int x^3 \ln x\,dx$.

(3)

(b) Explain why $\int_0^e x^3 \ln x\,dx$ is an improper integral.

(1)

(c) Evaluate $\int_0^e x^3 \ln x\,dx$, showing the limiting process used.

(3)
(Total 7 marks)

Q22.

(a) Explain why $\int_0^e \frac{\ln x}{\sqrt{x}}\,dx$ is an improper integral.

(1)

(b) Use integration by parts to find $\int x^{-\frac{1}{2}} \ln x\,dx$.

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(3)

(c) Show that $\int_0^e \frac{\ln x}{\sqrt{x}}\,dx$ exists and find its value.

(4)
(Total 8 marks)

Q23.

For each of the following improper integrals, find the value of the integral **or** explain briefly why it does not have a value:

(a) $\int_0^1 (x^{\frac{1}{3}} + x^{-\frac{1}{3}})\,dx$,

(4)

For more help, please visit our website www.exampaperspractice.co.uk

(b) $\int_0^1 \frac{x^{\frac{1}{3}} + x^{-\frac{1}{3}}}{x} \mathrm{d}x.$

(4)
(Total 8 marks)

Q24.

(a) Write down the value of

$$\lim_{x \to \infty} x\mathrm{e}^{-x}$$

(1)

(b) Use the substitution $u = x\mathrm{e}^{-x} + 1$ to find $\int \frac{\mathrm{e}^{-x}(1-x)}{x\mathrm{e}^{-x}+1} \mathrm{d}x.$

(2)

(c) Hence evaluate $\int_1^\infty \frac{1-x}{x+\mathrm{e}^x} \mathrm{d}x,$ showing the limiting process used.

(4)
(Total 7 marks)

Q25.

(a) For each of the following improper integrals, find the value of the integral **or** explain briefly why it does not have a value:

(i) $\int_0^9 \frac{1}{\sqrt{x}} \mathrm{d}x$;

(3)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(ii) $\int_0^9 \frac{1}{x\sqrt{x}} \mathrm{d}x$.

(3)

(b) Explain briefly why the integrals in part (a) are improper integrals.

(1)
(Total 7 marks)

Q26.

(a) Find $\int_0^a x\mathrm{e}^{-2x} \mathrm{d}x$, where $a > 0$.

(5)

(b) Write down the value of $\lim_{a \to \infty} a^k \mathrm{e}^{-2a}$, where k is a positive constant.

(1)

For more help, please visit our website www.exampaperspractice.co.uk

(c) Hence find $\int_0^{\infty} x\mathrm{e}^{-2x}\,\mathrm{d}x$.

(2)
(Total 8 marks)

Q27.

(a) Show that $\lim_{a \to \infty}\left(\frac{3a+2}{2a+3}\right) = \frac{3}{2}$

(2)

(b) Evaluate $\int_1^{\infty}\left(\frac{3}{3x+2} - \frac{2}{2x+3}\right)\mathrm{d}x$, giving your answer in the form 1n k, when k is a rational number.

(5)
(Total 7 marks)

Q28.

For each of the following improper integrals, find the value of the integral **or** explain briefly why it does not have a value:

(a) $\int_2^{\infty} 8x^{-3}\,\mathrm{d}x;$

(3)

(b) $\int_2^{\infty} \left(8x^{-3} + 1\right)\mathrm{d}x;$

(1)

(c) $\int_2^{\infty} 8x^{-3}(x+1)\,\mathrm{d}x.$

(3)
(Total 7 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

For more help, please visit our website www.exampaperspractice.co.uk